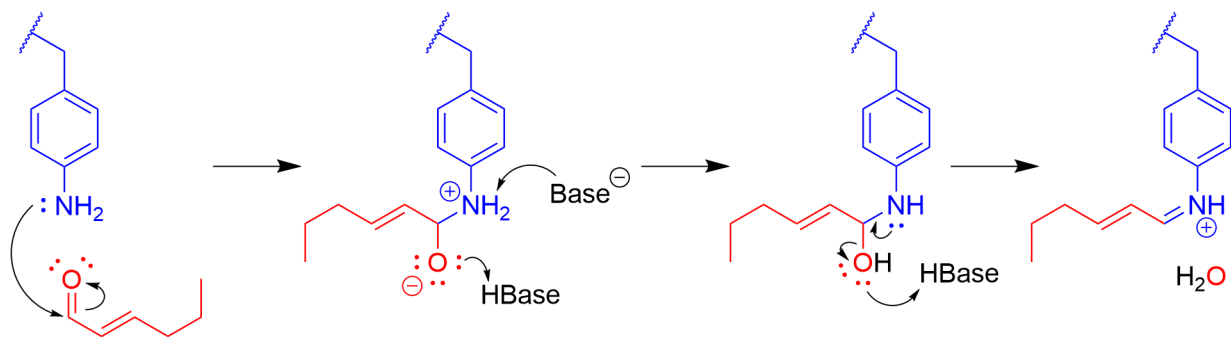


Step 1: Formation of activated iminium ion intermediate

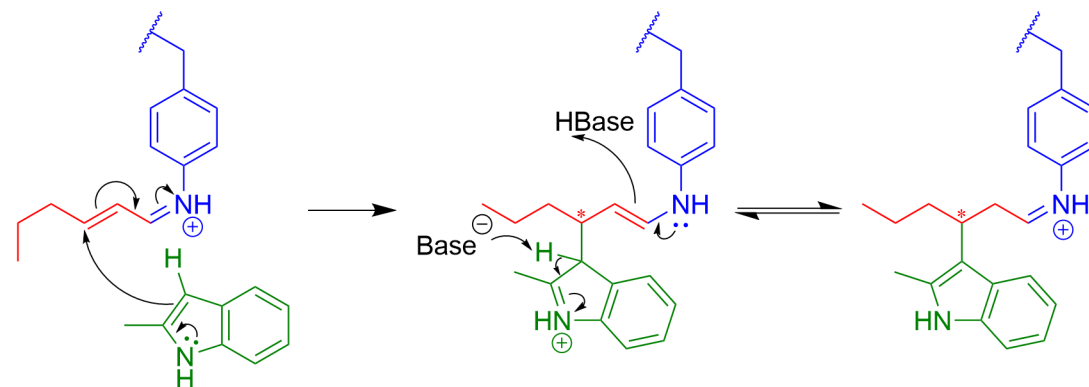


1.1 Nucleophilic attack of pAF's amino group at α -C of the aldehyde

1.2 Redistribution of charges/protons (HBase can be water or nearby residue)

1.3 Elimination of water

Step 2: Formation of adduct with indole substrate



Step 2.1: Formation of C-C sigma bond from pi-electrons

Step 2.2: Iminium-enamine tautomerization(s)

Step 3: Product release through hydrolysis (Step 1 in reverse)

